Training CNNs with Human Disagreement: Hard vs. Soft Label Regimes for Galaxy Morphology Classification

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***Abstract*—In the domain of astronomical imaging, it seems obvious that there may be disagreement among human scientists on the categorization of galaxies one billion light-years away. My primary question is whether a neural network trained against vote-fraction distributions (human disagreement data) produces better-calibrated morphology classifiers than training against majority-vote labels. I trained two identically-configured convolutional neural networks (CNNs) on $\sim$35k GalaxyZoo2 images, one with cross-entropy loss on hard labels and one with KL divergence on soft branch-product targets derived from the initial vote fractions. Testing over 30 epochs showed soft-label training improved top-1 accuracy by 3.4% overall, but produces a directional asymmetry: ellipticals become more accurate (+10%) while face-on non-spirals are less accurate (-9%). Per-class analysis reveals that soft-label training produces large calibration gaps (specifically, underconfidence) on classes where it is most accurate, and better calibration only on the class where it is not. KL divergence from vote fractions is substantially lower for the soft-label model. The results indicates that soft-label training improves accuracy at one specific class boundary while producing systematic underconfidence across its high-accuracy classes.**

I. Introduction

Galaxy morphology classification is a long-standing astronomical task, as understanding the distribution of galaxy shapes and structure informs cosmological models, current theories on galactic evolution, and research on large-scale galactic structures. Modern surveys like the Sloan Digital Sky Survey (SDSS) image hundreds of millions of galaxies, making automated classification and analysis an essential development for the field at large.

With the average distance of a galaxy in the SDSS being $>$2 billion light-years, the images of these galaxies are prone to artifacting, low signal-to-noise ratios, occlusion, and limits on image resolution. As such, human scientists can easily disagree on what features are present in the image and into which class the galaxy may fall. This disagreement is not a failure of the annotation process, rather it reflects real physical ambiguity in what the image depicts, especially at moderate redshift or limited angular resolution. GalaxyZoo2 (GZ2) addresses this by having many citizen scientists annotate each galaxy and recording the distribution of their answer (vote fractions) rather than forcing a single label [1]. The data consists of roughly 240k galaxies from SDSS, each with per-question vote fractions along a morphological decision tree. This produces a rich dataset that encapsulates both a plausible majority classification and a measure of collective human confidence.

Standard practice in galaxy classification has been to use one-hot labels derived from the majority vote. This methodology is easily justified, but is not a requirement. Loss functions exist that allow a network to train against a full target distribution, such as Kullback-Leibler (KL) divergence or soft cross-entropy. In principle, if we instead use the raw vote fractions as the training target, the model may learn not just which class is most likely but also how confident to be in that classification. The primary question is to determine if such a training regime produces a meaningful change in the performance and decision-making of a model when compared to traditional single-label training.

The approach was to train two identically-configured CNNs on a simplified four-class morphology set, one with cross-entropy (CE) against majority-vote labels and one with KL divergence against vote-fraction targets. Both were evaluated on accuracy, expected calibration error, distributional agreement with vote fractions, and attention patterns. Soft-label training produced measurable shifts in model behavior, but those shifts were directional and class-dependent rather than a uniform improvement: the soft-label model shows a directional shift at the elliptical/face-on non-spiral boundary, im-

proving elliptical accuracy while reducing accuracy on morphologically similar non-spirals. The model also appears to become systematically underconfident on classes where it is accurate, while remaining well-calibrated on classes where it is not. This asymmetry is consistent with a mechanism by which soft targets encode inter-class similarity, though confounds discussed in Section V-C limit me from making a confident causal interpretation.

II. Related Work

This project draws upon three lines of prior research: convolutional networks for galaxy morphology classification, training against non-one-hot targets derived from human disagreement, and calibration in modern neural networks.

Dieleman et al. [2] established the feasibility of CNN-based galaxy morphology classification on the GZ2 corpus. Their rotation-invariant network won the 2014 Kaggle Galaxy Challenge by predicting all 37 vote fractions in the GZ2 decision tree as independent regression targets. This framing of vote fractions as real-valued regression outputs accurately reproduces vote fractions in aggregate, but does not produce a normalized probability distribution over disparate morphological classes. Consequently, much of the prior CNN work on GZ2 has not directly compared model confidence to the shape of human disagreement on individual galaxies. I adopt a classification framing with four collapsed classes and instead use vote fractions as soft targets, enabling per-image distributional comparison between model output and human vote distribution; this comparison is not enabled by the regression-based framing of prior work.

Beyond domain-specific work, a broader literature has examined training against non-one-hot targets in image classification. Peterson et al. [3] demonstrated on CIFAR-10H, a variant of CIFAR-10 using distributions of labels from human annotators, that replacing one-hot labels with human-derived distributions during training improves generalization, adversarial robustness, and calibration. Their central claim is that human disagreement represents a useful training signal, rather than noise to be normalized. This motivates my core hypothesis, however their evaluation focuses on aggregate metrics on data with relatively sharp class boundaries. I apply the same hypothesis to the GZ2 dataset, a domain with naturally ambiguous morphological boundaries, and add ambiguity-stratified evaluation to expose effects not otherwise visible in aggregate metrics.

Calibration itself has a well-established measurement framework. Guo et al. [4] rigorously document calibration metrics and establish that modern CNNs are systematically overconfident. Their work introduces Expected Calibration Error (ECE) as a standard calibration metric and reliability diagrams as standard diagnostic tools. I adopt this methodology as the primary metric for model comparison, allowing for a more introspective quantitative measurement of network confidence and calibration. The finding that modern CNNs are typically overconfident sets the expectation that soft-label training should reduce ECE relative to a CE baseline. As I report in Section IV-C, the CE baseline is already well-calibrated in aggregate, which changes the baseline expectation and interacts with the calibration effects I observed.

III. Methods

A. Dataset

The main GalaxyZoo2 [1] image corpus comprises roughly 240k 424×424 RGB JPEGs. I merged three CSVs: `gz2_hart16.csv` (debiased vote fractions), `gz2sample.csv` (redshift and photometric metadata), and `gz2_filename_mapping.csv` (map of object IDs to image assets). The vote fraction data used is that released by Hart et al. [5], which corrected redshift-dependent bias on vote fractions across the decision tree.

Images were center-cropped from 424×424 to 224×224, facilitated by the analysis of metadata indicating the 90th percentile of pictured galaxy structures spanned ∼86-130 pixels across, meaning a center crop could be performed to trim the dominant sky background while avoiding the information loss associated with downsampling.

Class separation collapsed the 37-node GZ2 decision tree, seen in Figure 1, to four terminal classes: Elliptical, Edge-on disk, Face-on spiral, Face-on non-spiral. This set of classes is chosen as they are defined by the presence of detectable features rather than their absence, producing a more coherent vote-fraction label than, say, including lenticular galaxies.

Elliptical galaxies are indicated by an answer of 'smooth' (A0) on task 0 of the GZ2 decision tree. Edge-on disk is characterized by an answer of 'yes' on task 1, and face-on spiral is separated from face-on non-spiral by an answer of 'spiral' or 'no spiral' on task 3, respectively.

Hard labels were constructed using a confidence threshold of 0.7 on each relevant decision-tree branch. Rows that did not meet the threshold are discarded as overly ambiguous. After filtering, ∼178k galaxies remain, subsampled to 50,000 for this project (stratified by class to preserve natural proportions: ∼40% Elliptical,

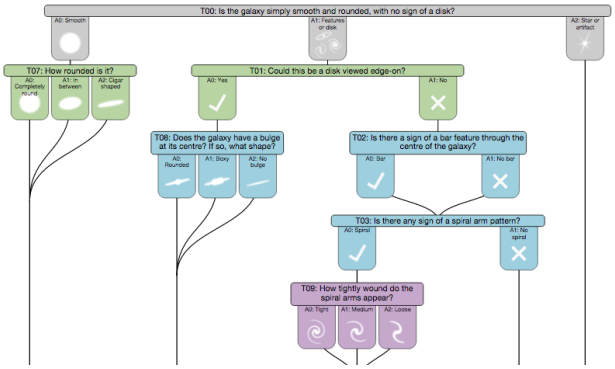


Fig. 1: Truncated upper-half of GZ2 decision tree.

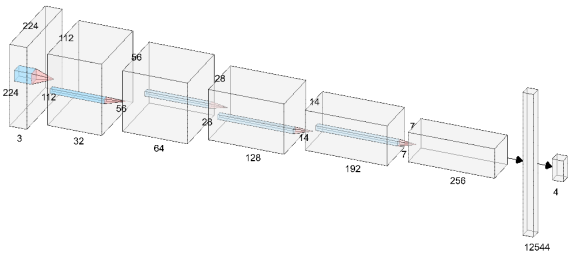


Fig. 2: CNN Architecture

$\sim$45% Face-on spiral, $\sim$7% Edge-on disk, and $\sim$8% Face-on non-spiral).

Soft target construction used the branch-product from the decision tree:

- P(Elliptical) = p(smooth)
- P(Edge-on) = p(featured) $\times$ p(edge-on)
- P(Face-on spiral) = p(featured) $\times$ p(not edge-on) $\times$ p(spiral)
- P(Face-on non-spiral) = p(featured) $\times$ p(not edge-on) $\times$ p(no spiral)

These were re-normalized per-row to sum to 1. NaN vote fractions in unreached branches were zeroed before calculating branch-product. This retained a $\sim$99% agreement rate between argmaxed soft distribution and hard label.

Data was stratified with a 70/15/15 split for training, validation, and testing, respectively. Identical splits were used for both training regimes.

Class imbalance was handled by deriving per-sample weights, assigned in inverse-proportion to class frequency, and used via `WeightedRandomSampler` during training. Validation and test loaders used natural class distribution.

Normalization was done via per-channel (RGB) mean and standard deviation, calculated from a subset of 10,000 training images and applied identically across training, validation, and testing sets.

$$\mu \approx [0.045, 0.039, 0.029], \quad \sigma \approx [0.088, 0.073, 0.063]$$

The low mean and standard deviation reflect the dominance of the dark sky over the total image intensity, even after cropping.

B. Architecture

The architecture is a modest, compact CNN designed around three principles: (1) sufficient capacity to learn the task while still allowing behavioral variation to be attributable to the loss function, (2) GAP + linear classifier to reduce parameter count and enable Grad-CAM interpretation, and (3) simplicity of the forward pass (no skip connections) to further ensure results can be attributed to the loss function.

The architecture consists of a 7×7 stride-2 stem (32 channels), followed by four blocks of increasing channel depth (64, 128, 192, 256), each containing one stride-1 and one stride-2 convolution layer, the latter reducing spatial size by a factor of $\frac{1}{2}$. Output feeds into global average pooling, a dropout layer ($p = 0.4$), and a linear layer to 4 output logits. Each convolution layer is followed by BatchNorm, each block uses ReLU as activation, and weights are initialized using Kaiming initialization.

C. Training

Both models share identical training configuration except for the loss function, enabling direct comparison of loss-driven effects. Training used the AdamW optimizer with initial learning rate 3×10^{-4} and weight decay 1×10^{-4}. Learning rate was scheduled using cosine annealing, configured to decay to zero over 30 epochs. Batch size was fixed at 128 images throughout.

Training-set images were augmented with random rotations up to 180° followed by random, independent horizontal and vertical flips ($p = 0.5$ each), applied prior to center cropping to trim rotation artifacts. These augmentations are principally motivated by the natural rotation and reflection invariance of galaxies in images. Validation and testing did not include these augmentations.

Training used a `WeightedRandomSampler` with per-sample weights (derived in Section III-A) to overcome class imbalance; validation and testing used the natural class distribution.

The hard-label regime uses standard categorical cross-entropy:

$$\mathcal{L}_{CE} = -\log p_y \tag{1}$$

where p_y is the model's softmax probability on the true class index y.

The soft-label regime uses soft-target cross-entropy:

$$\mathcal{L}_{KL} = -\sum_{k=1}^{K} q_k \log p_k \tag{2}$$

where $q = (q_1 ... q_k)$ is the branch-product soft target defined in Section III-A and p is the model's softmax output. This objective is equivalent to minimizing Kullback-Leibler (KL) divergence $\mathrm{KL}(q||p)$ up to a target-only constant, meaning the gradient with respect to the model parameters is identical.

The soft-target loss was implemented explicitly rather than using PyTorch's `F.kl_div`, as the latter requires log-probability inputs, internally evaluating $q \log q$, which produces NaN when q contains exact 0.0 values. The explicit formulation only evaluates $\log p$, which is always strictly positive after softmax.

All randomization-related seeds (model initialization, data splitting, and DataLoader workers) were fixed to 112568 (chosen randomly). Both regimes used the same seed, same data split, and same initial model weights, only differing in loss function. Training was done on a single NVIDIA RTX 5070Ti and took approximately 30 minutes per 30-epoch run. During training, two checkpoints were saved: the epoch with the lowest validation loss, and the final epoch.

D. Evaluation Metrics

1) Classification Metrics: I report top-1 accuracy (fraction of predictions matching the hard label), per-class accuracy (top-1 accuracy within each class), and a 4×4 confusion matrix with both raw counts and row-normalized proportions.

2) Expected Calibration Error (ECE): ECE measures the average gap between predicted confidence and actual accuracy. This is done by partitioning predictions into M equal-width bins based on softmax confidence, then computing (a) the mean predicted confidence and (b) the actual prediction accuracy in each bin. ECE is the weighted mean of these absolute differences:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \cdot |\mathrm{acc}(B_m) - \mathrm{conf}(B_m)| \tag{3}$$

where B_m is the set of predictions in bin m, $|B_m|$ is its size, and N is the total size of the dataset. For my evaluation, ECE was computed using $M = 10$.

3) Reliability Diagram: Visualization accompanying ECE that plots mean confidence (x-axis) against actual accuracy (y-axis) for each bin. Perfect calibration traces the diagonal $y = x$.

4) Brier Score: Mean squared error between full predicted probability distribution and the one-hot encoded true label.

$$\mathrm{BS} = \frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{K} (p_{i,k} - 1[y_i = k])^2 \tag{4}$$

where N is the total sample count and K is the number of classes. Unlike ECE, this score is sensitive to predictions across the full distribution, not just the argmax, and can better indicate performance on ambiguous samples that ECE would undervalue.

5) Average KL Divergence: Across all test samples, compute the average of all $\mathrm{KL}(q_i||p_i)$:

$$\mathrm{KL} = \sum_{k=1}^{K} q_{i,k} \log(q_{i,k}/p_{i,k}) \tag{5}$$

$$\overline{\mathrm{KL}} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{KL}(q_i||p_i) \tag{6}$$

6) Ambiguity Stratification: Each of the metrics described above were computed separately on two additional test subsets:

- Clean: examples where $\max_k q_{i,k} \geq 0.8$ (vote fraction concentrated on one class)
- Ambiguous: examples where $\max_k q_{i,k} < 0.8$ (vote fraction spread across multiple classes)

This was done to isolate whether the soft-target training allowed the model to carry uncertainty information and provide a meaningful difference. The cutoff of 0.8 was chosen due to it splitting the test dataset into roughly 72% clean/28% ambiguous subsets, allowing for each to be of reasonable size while still being semantically meaningful as "ambiguous or not".

IV. Experiments and Results

A. Experimental Setup

I evaluated two models trained with the setup described in Section III, differing only in the loss function used: one with cross-entropy loss against hard-label targets and the other using the modified Kullback-Leibler loss (2) against soft-label targets. For this section, these models will be referred to as "CE" and "KL", respectively. Both were evaluated on the held-out test set of 7,500 images using the best validation checkpoint from each 30-epoch training run.

B. Classification Performance

1) Overall Performance: Across the entirety of the test set, CE showed a top-1 accuracy of 80.4%, and KL showed 83.8%, a 3.4 percentage point improvement by moving to soft-label training.

2) Per-Class Accuracy: The accuracy improvement seen by KL was not uniform across all classes.

While edge-on disk and face-on spiral classes show minimal differences in accuracy, elliptical and face-on non-spiral accuracies shifted substantially in opposite directions.

3) Confusion Matrices: Figure 3 shows the confusion matrices for both the CE and KL models.

The shift in performance is concentrated at the boundary between elliptical galaxies and face-on non-spiral galaxies, with the highest degree of confusion for both models being between these two classes. CE misclassifies 16% of ellipticals as face-on non-spirals, which KL reduces to only 7.0%. Conversely, KL misclassifies 35% of face-on non-spirals as ellipticals while CE misclassifies only 22%. Other class confusions are far more minimal, and see little change between the models.

C. Calibration

1) Overall Calibration: At the aggregate level, CE reports an ECE of 0.026, while KL gave 0.103, a noteworthy degradation in calibration when moving to soft targets.

Brier score for both models was nearly tied, with CE reporting 0.265 and KL 0.258. The nearly-identical Brier scores despite a significant ECE gap reflects the fact that Brier scores measure the distribution quality among all classes, rather than ECE's measurement of the max-confidence predictions.

2) Stratified Calibration: Tables II, III, and IV depict model accuracy, ECE, and Brier Scores across the three strata, with better scores highlighted per-stratum for each class.

Per-class stratified ECE reveals that the aggregate reversal (CE better on clean, KL better on ambiguous) is not uniform. KL is more poorly-calibrated than CE across all strata on Elliptical and Edge-on disk, and better-calibrated across all strata on Face-on non-spiral. Only Face-on spiral shows the stratum reversal which, combined with class-composition differences between strata, drives the aggregate-level pattern. A secondary pattern also emerges: KL's per-class ECE correlates with its per-class accuracy, being highest on Ellipticals (0.199 at 90.5% accuracy) and Edge-on disk (0.176 at 93.2%), and lowest on Face-on non-spirals (0.118 at 52.9%). CE's ECE, by comparison, is uniformly low across classes.

3) Reliability Diagrams: Figure 4 shows reliability diagrams for each model. CE's predictions track the diagonal closely across the confidence range, consistent with its aggregate ECE of 0.026. KL's predictions rise systematically above the diagonal across every confidence bin, indicating systematic underconfidence: the model's reported confidence underestimates its actual accuracy, with performance exceeding reported confidence by 8-18% across the mid-confidence bins.

D. Distributional Agreement

Table V reports the mean per-sample KL divergence between each model's softmax output and the ground-

TABLE I: Per-Class Performance

	Model Accuracy (%)		
Class	**CE**	**KL**	**Change (%)**
Elliptical	80.76	90.50	+9.74
Edge-on disk	95.22	93.20	-2.02
Face-on spiral	80.81	81.64	+0.83
Face-on non-spiral	62.21	52.90	-9.31

TABLE II: Model Accuracy by Class and Stratum

	Overall		Clean		Ambiguous	
Class	CE	KL	CE	KL	CE	KL
Elliptical	80.8	**90.5**	82.2	**92.5**	75.8	**83.7**
Edge-on disk	**95.2**	93.2	**99.6**	99.2	**91.7**	88.4
Face-on spiral	80.8	**81.6**	88.1	**89.1**	55.4	**55.8**
Face-on non-spiral	**62.2**	52.9	**72.2**	65.1	**56.3**	45.7
Overall	80.4	**83.8**	85.5	**90.0**	67.4	**67.8**

*Higher accuracy is better.

TABLE III: Model ECE by Class and Stratum

	Overall		Clean		Ambiguous	
Class	CE	KL	CE	KL	CE	KL
Elliptical	**0.024**	0.199	**0.022**	0.205	**0.052**	0.178
Edge-on disk	**0.011**	0.176	**0.015**	0.150	**0.019**	0.201
Face-on spiral	**0.048**	0.062	**0.039**	0.067	0.187	**0.089**
Face-on non-spiral	0.144	**0.118**	0.123	**0.080**	0.174	**0.168**
Overall	**0.026**	0.103	**0.023**	0.126	0.098	**0.045**

*Lower ECE is better.

TABLE IV: Brier Scores by Model and Strata

	Brier Score		
Model	**Overall**	**Clean**	**Ambiguous**
CE	0.2653	0.1933	0.4510
KL	**0.2582**	**0.1832**	0.4516

*Lower Brier score is better.

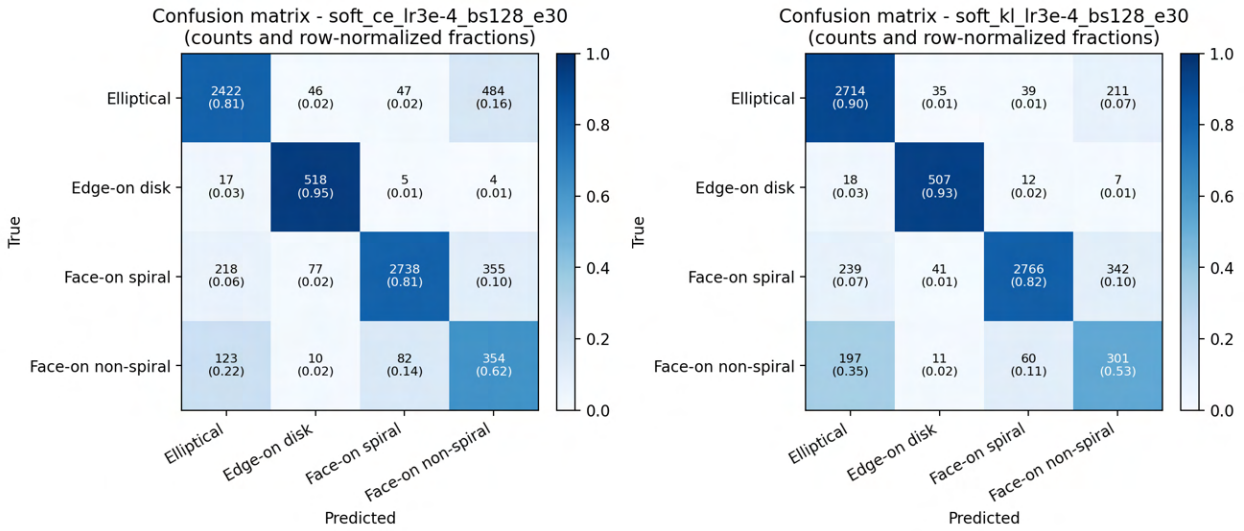


Fig. 3: Confusion matrices for CE (left) and KL (right) across all 4 classes

truth vote fraction distribution.

TABLE V: Mean KL$(q\|p)$ by Model and Stratum

Model	Overall	Clean	Ambiguous
CE	0.478	0.358	0.787
KL	**0.281**	**0.249**	**0.365**

*Lower KL divergence is better.

KL achieves lower divergence from vote fractions than CE across all strata. The advantage is largest along the ambiguous stratum (0.787 vs. 0.365, a $\sim$54% reduction) and smallest on clean examples (0.358 vs. 0.249, a $\sim$30% reduction). On clean examples, both models produce near-one-hot distributions, limiting the room for noteworthy distributional differences; on ambiguous examples, vote fractions are genuinely spread across multiple classes, and KL's predictions track this spread significantly more closely than CE's.

This result is the expected direction for soft-label training, as the loss function explicitly minimizes the divergence from vote fractions during training, and as such should be expected to match those fractions more closely than a model trained on one-hot hard labels.

E. Qualitative Attention Analysis

To complement the quantitative findings, I generated Grad-CAM attention maps for both models on a curated set of test examples. Attention was computed with respect to the final convolution block ($7 \times 7 \times 256$ feature map) using each model's own predicted class as the target. Heatmaps were upsampled bilinearly to 224×224 and overlaid onto the input image. Figure 5 shows four test examples I selected to illustrate particular behavioral differences. Row 1 shows an Elliptical galaxy that CE misclassified as Face-on non-spiral, but KL identified correctly, despite each having similar attention regions. KL's off-center attention pattern may indicate a subtle feature variation leading to its correct classification. Row 2 shows the inverse result (CE correct, KL incorrect), with KL's attention spread to the outer regions vs. CE's attention on the entire luminous region. Row 3 shows an unambiguous Elliptical galaxy correctly classified by both models, with broadly similar attention patterns. Row 4 shows a galaxy with a highly-ambiguous soft target ([.02, 0.0, 0.491, 0.493]) that both models correctly classify with significantly different attention patterns.

V. Discussion

A. Class-Boundary Effects

The most striking finding from the comparison between the two models was that nearly all behavioral differences were concentrated at a single class boundary: that between Elliptical and Face-on non-spiral galaxies. Misclassification was most prevalent along these two classes for both models, with minimal changes in accuracy among the other two classes. The aggregate 3.4% accuracy improvement from KL's training is therefore the net balance of a large class-specific shift, not a uniform improvement across the task as I initially thought.

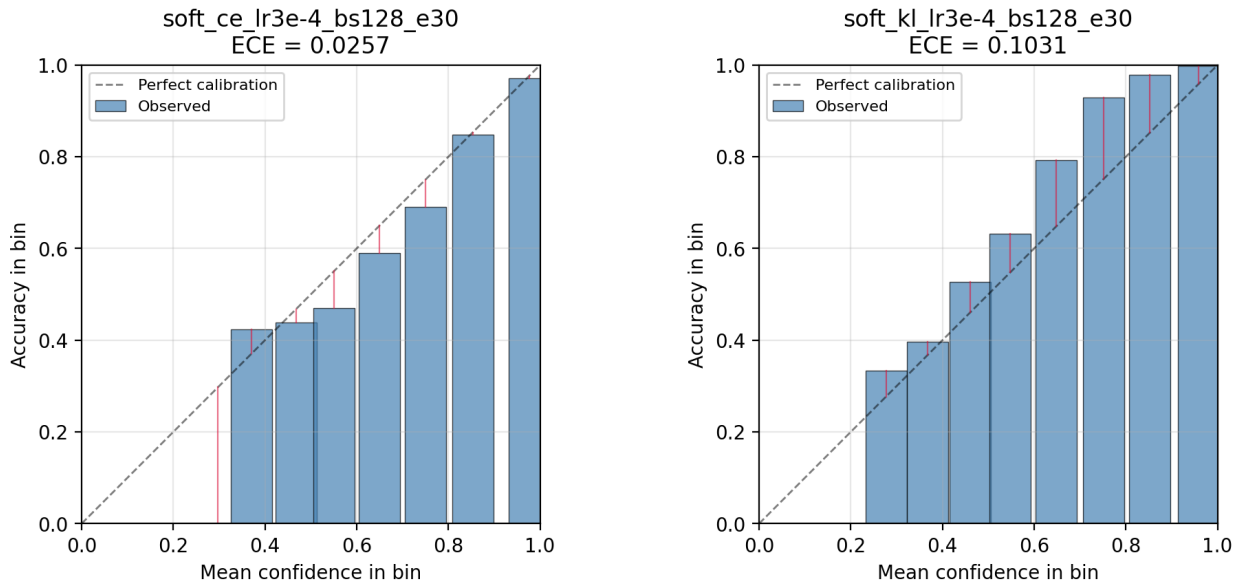


Fig. 4: Reliability diagrams for CE (left) and KL (right) models

This localization suggests that the choice of loss function may not act uniformly on the decision boundary of a CNN classifier; rather, it affects specific regions of class space asymmetrically. In this case, that region is the Elliptical/Face-on non-spiral transition, comprised of two classes whose morphological distinction is genuinely subtle, even to human observers, and whose class frequencies differ substantially (ellipticals are roughly five times more prevalent than Face-on non-spirals in this set).

I propose that the directional shift reflects the soft targets encoding of inter-class similarity by placing more predictive weight on morphologically-adjacent classes, and that the model learns a decision boundary that favors the more common class across that boundary. This conclusion is not fully substantiated by the current data, as I discuss in Section V-C.

B. Calibration and Per-Class Accuracy

A second substantial pattern appears in the calibration analysis: under KL training, the classes with the highest test accuracy also have the largest calibration gaps, and the class with the lowest KL accuracy has the most well-behaved calibration under KL. Under hard label training, per-class ECE for CE is low to moderate across all classes, showing no correlation with accuracy.

This pattern is consistent with a mechanism in which soft-label training teaches the model to distribute probability mass across morphologically adjacent classes, reflecting the structure of the soft targets themselves. On elliptical samples, for instance, KL's softmax outputs place most mass on Elliptical, but reserve some for the morphologically-adjacent Face-on non-spiral class, yielding typical max-class confidences around 0.75-0.85 rather than the near-unity values CE tends to produce. For Edge-on disk, the mechanism is less clear; morphological adjacency alone cannot explain the high ECE, and the pattern may reflect more general effects of soft-label training on confidence calibration that require further investigation.

CE's small ECE reflects a model whose confidence tracks closely with its accuracy, while KL's aggregate ECE of 0.103 reflects a model that is too cautious about its own reliability. Whether this is a desirable property depends on the downstream use of the confidence estimates.

The aggregate stratum-level reversal described in Section IV-C is similarly not a fundamental property of soft-label training versus hard-label training on the data, but a compositional artifact that per-class evaluation reveals. My initial conclusion upon seeing the aggregate-stratified ECE was that soft-label targets calibrate to ambiguous examples better. While this is narratively cleaner, the per-class stratification has moved me away from this, and shows itself to be a necessary tool for distinguishing genuine effects from composition-driven artifacts.

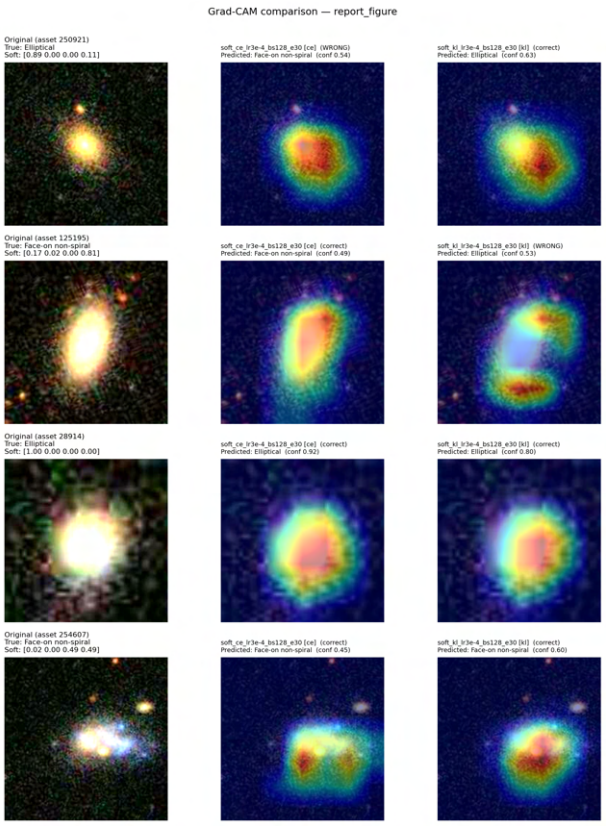


Fig. 5: Grad-CAM attention map, with each row showing the original image, CE's attention mask, and KL's attention mask

C. Alternative Interpretations and Limitations

Two confounds limit the strength of the proposed mechanism. First, the branch-product soft-target construction imposes a natural asymmetry independent of morphological similarity: Elliptical targets are single vote fractions while Face-on non-spiral targets are products of three factors and are mathematically more spread, by construction. A model trained on these targets may learn concentrated Elliptical and spread Face-on non-spiral predictions via arithmetic alone, although this effect is not seen in the other two classes — Face-on spiral shares the same 3-factor construction without the same underconfidence pattern. Distinction between the morphological similarities and artifacts from the decision tree may require comparison with an alternative soft-target construction (e.g. label smoothing). Second, the analysis was done with a constant random seed for consistent introspection/reproducibility. Per-class metrics on smaller strata may reflect sampling variation rather than robust effects; multi-seed replication would establish which patterns are genuinely reproducible.

D. Summary and Future Work

This research sought to compare hard-label cross-entropy and soft-label Kullback-Leibler training on a simplified galaxy morphology task, using identical architecture, hyperparameters, and data splits for each regime, differing only by loss function.

The principal observations are: (1) soft-label training produced a 3.4% overall accuracy improvement; (2) this improvement was concentrated almost entirely at a single class boundary; (3) the soft-label model produced output distributions that matched vote fractions more closely than hard labels across all strata; (4) soft-label training produced systematic underconfidence on classes where it achieved high accuracy, leading to large ECE gaps despite the model's stronger prediction quality; and (5) aggregate-level stratum ECE reversals are likely compositional artifacts driven by class behavior and distribution within data.

Several directions for future work may be inspired by the confounds in this project. Comparison with alternative soft-target constructions (e.g. Dirichlet-smoothed vote fractions, label smoothing) would separate decision-tree-depth effects from genuine morphological similarity effects, and distinguish vote-fraction-specific effects from general non-one-hot training effects. Multi-seed replication would establish the robustness of the finer-grained per-class per-stratum patterns reported here. Evaluation on class-balanced morphological data would probe the frequency-dependence of the directional shift.

Code can be seen and explored at https://github.com/theonlynanu/GalaxyZooClassification

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